Graphing Sine and Cosine Functions

Name _____

Algebra 2 Lesson 7-3 (fast track) Date _____ Period _____

Learning goal: I can find the amplitude, period and frequency of a sine or cosine function from its equation and graph it from five key points.

Language goal: I can explain how a and b change the graph of $y = a\sin bx$ using the words *amplitude*, *period*, *midline*, and *frequency*.

On the quiz: (1) Amplitude and period: Example 3. (2) Rate of change from key points: Example 4, Practice. (3) Equation against a graph (full Savvas quiz only). (4) Reflect $5\cos(2x)$: Example 3B, Try It 4b. (5) Tide (full Savvas quiz only).

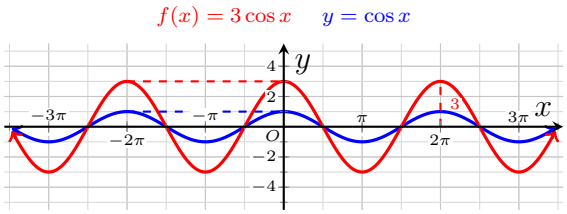
On the Topic 7 test: Question 17 asks for a frequency (Example 3); 11 and 18 ask for an average rate of change (Example 4). Questions 2 (equation against a graph), 6 (tide) and 12 (repeating sine values) come from Examples 6, 5 and 1, which this packet skips.

Example 3: Identify Amplitude and Period

For $y = a\sin bx$ or $y = a\cos bx$ ($a \neq 0$, $b \neq 0$): the **amplitude** is $|a|$ (the distance from the **midline** to a maximum or minimum) and the **period** is $\frac{2\pi}{|b|}$ (the length of one cycle).

A. What are the amplitude and period of $f(x) = 3\cos x$?

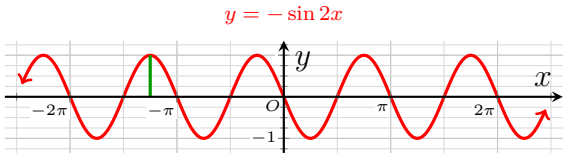
1. Graph $f(x) = 3\cos x$ and the parent $y = \cos x$.
2. Midline $y = 0$. Maximum values 3, minimum values -3: 3 times those of $\cos x$.
3. The amplitude of $f(x) = 3\cos x$ is 3, and the period is 2π.



Amplitude 3, period 2π.

B. What are the amplitude and period of $y = -\sin 2x$?

1. $y = -f(x)$ is a **reflection** of the parent $y = \sin x$ over the x-axis.
2. $y = f(2x)$ is a horizontal compression by a factor of $\frac{1}{2}$.
3. Neither one changes the amplitude: $|-1| = 1$.
4. Period compressed by $\frac{1}{2}$: $\frac{1}{2}(2\pi) = \pi$.
5. Check with a graph:



Repeats every π. **Amplitude 1, period π.**

Say it: The amplitude of $y = -\sin 2x$ is _____ because the midline is _____ and a maximum is _____ above it. The period is _____ because

Try It 3: What are the amplitude and period of each function?

a. $y = \frac{1}{3}\cos\frac{1}{2}x$

amplitude: _____ period: _____

b. $y = 2\sin \pi x$

amplitude: _____ period: _____

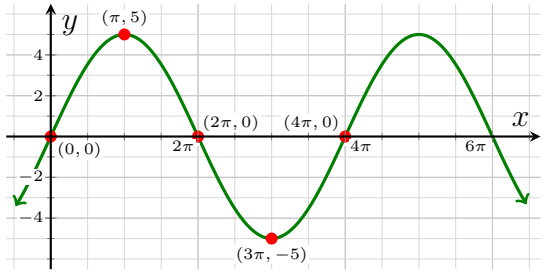
Frequency is the reciprocal of the period: the number of complete cycles in one unit of x. $y = \sin x$ and $y = \cos x$ have period 2π and frequency $\frac{1}{2\pi}$; $y = -\sin 2x$ has period π and frequency $\frac{1}{\pi}$.

Example 4: Graph $y = a\sin bx$ and $y = a\cos bx$

A. Graph $f(x) = 5\sin\frac{1}{2}x$.

1. $a = 5$: vertical stretch, so the amplitude is 5.
2. $b = \frac{1}{2}$: horizontal stretch by a factor of 2, so the period is $\frac{2\pi}{1/2} = 4\pi$.
3. Divide one period into fourths: $\frac{4\pi}{4} = \pi$. Plot, in order, a zero, a maximum, a zero, a minimum and a zero.

$f(x) = 5\sin\frac{1}{2}x$

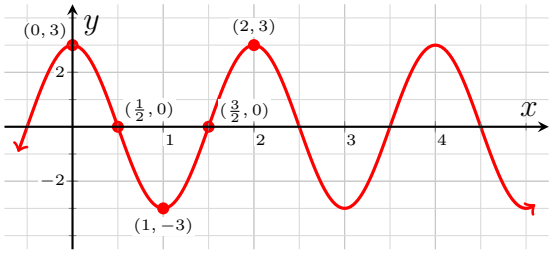


Domain $(-\infty, \infty)$, range $[-5, 5]$. **Amplitude 5, period 4π, frequency $\frac{1}{4\pi}$.**

B. Graph $f(x) = 3\cos\pi x$.

1. $a = 3$: vertical stretch by a factor of 3, so the amplitude is 3.
2. $b = \pi$: horizontal compression by a factor of $\frac{1}{\pi}$, so the period is $\frac{2\pi}{\pi} = 2$.
3. Fourths: $\frac{2}{4} = \frac{1}{2}$. Plot, in order, a maximum, a zero, a minimum, a zero and a maximum.

$f(x) = 3\cos\pi x$



Domain $(-\infty, \infty)$, range $[-3, 3]$. **Amplitude 3, period 2, frequency $\frac{1}{2}$.**

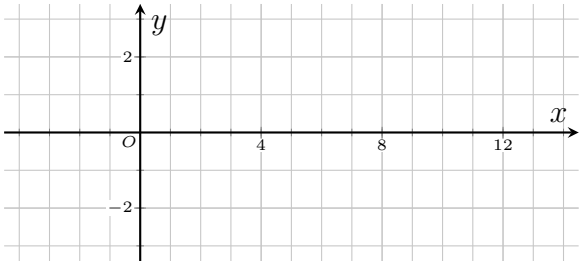
When $|b| < 1$ the graph is stretched horizontally and the period is longer than 2π; when $|b| > 1$ it is compressed and the period is shorter.

Say it: In $f(x) = 5\sin\frac{1}{2}x$, the amplitude is _____ because $a =$ _____. The period is _____ because $2\pi \div$ _____ = _____, so the frequency is _____ and the key points are _____ apart, starting on the _____ $y = 0$.

Try It 4: Graph the functions. What are the key features?

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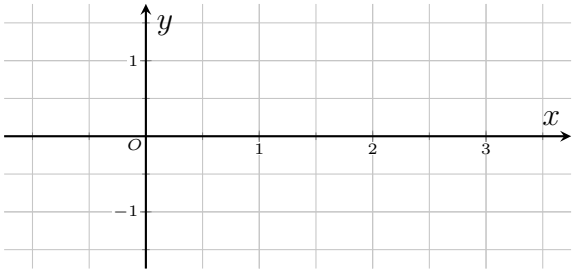
a. $g(x) = \frac{3}{2}\cos\frac{3}{4}x$



amplitude _____ period _____

frequency _____ range _____

b. $h(x) = -\sin 2\pi x$



amplitude _____ period _____

frequency _____ range _____

Practice: Average rate of change from the key points

Use the key points you plotted in Example 4.

Hint: the average rate of change of f on the interval $[p, q]$ is $\dfrac{f(q) - f(p)}{q - p}$.

a. What is the average rate of change of $f(x) = 5 \sin \frac{1}{2}x$ on the interval $[0, \pi]$?

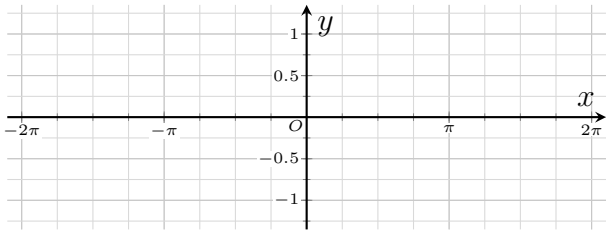
b. What is the average rate of change of $f(x) = 3 \cos \pi x$ on the interval $[0, 1]$?

Extension: DOK 3, Performance Task (Practice 23)

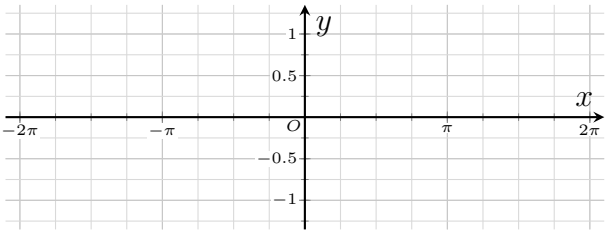
Self-paced; from Example 6, which this packet skips. Parts A and C use Example 4's five key points.

Dominga is investigating how the signs of the parameters a and b create transformations of the sine function.

Part A Graph $f(x) = \sin 2x$ and $g(x) = -\sin(2x)$ on the same coordinate plane.



Part C Graph $f(x) = \sin(2x)$ and $h(x) = \sin(-2x)$ on the same coordinate plane.



Part B How are the graphs of $f(x) = \sin(2x)$ and $g(x) = -\sin(2x)$ related?

Part D How are the graphs of $f(x) = \sin 2x$ and $h(x) = \sin(-2x)$ related?

Hint: for Parts A and C, one period is $\frac{2\pi}{2} = \pi$. Divide it into fourths and plot a zero, a maximum or minimum, a zero, the other one, and a zero.

Part E How is the graph of $y = a \sin(bx)$ affected when a or b is replaced with its opposite? Explain.

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
6H8	Evaluate trigonometric ratios using reference angles: sin, cos, tan	Warm-up: the sine and cosine values at the key points (Lesson 7-2)
9S8 ★	Find properties of sine and cosine functions	Quiz 1: amplitude $\lvert a\rvert$ and period $2\pi/\lvert b\rvert$ from the equation (Example 3); Quiz 4 bonus: a negative a keeps the amplitude
A7V ★	Graph sine and cosine functions	Quiz 2: the five key points of $y = a\sin bx$ (Example 4); Quiz 4 bonus: graph a reflection over the x-axis
5XN	Write equations of sine and cosine functions from graphs	Quiz 3 (full Savvas quiz only): amplitude and period read off a graph

Before the quiz, finish these sentences. (Spanish: amplitude = *amplitud*, period = *período*, midline = *línea media*, frequency = *frecuencia*, periodic function = *función periódica*.)

The _____ is the distance from the midline to a maximum or a minimum, so the amplitude of $y = a\cos bx$ is _____.

The _____ is the length of one cycle, so the period of $y = a\sin bx$ is _____.

The _____ is the reciprocal of the period, so a function with period 4π has frequency _____.

The five key points of $y = a\sin bx$ are _____ of a period apart, and the first one is on the _____.

Changing a to $-a$ does not change the amplitude; it _____.

Exit: Explain how a and b change the graph of $y = 4\cos 2x$. The _____ is _____ because $a = 4$, and the _____ is _____ because

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Its endpoint values on $\left[0, \frac{\pi}{2}\right]$ are $(0,$ _____$)$ and $\left(\frac{\pi}{2},$ _____$\right)$, so its average rate of change on $\left[0, \frac{\pi}{2}\right]$ is _____.